

// Day 18 of 30

BEGINNER-FRIENDLY GUIDE

Docker Explained — From Zero to Deploy

→ Swipe through all 6 slides



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

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What Problem Docker Solves

It works on MY machine...

Without Docker:

Dev uses Java 17, MySQL 8
Server has Java 11, MySQL 5
→ App crashes in production

With Docker:

Package app + ALL dependencies
→ Same everywhere
→ Dev = Staging = Production

💡 Docker = ship your entire environment



Shubham Bhati

linkedin.com/in/bhatishubham

Image vs Container

Blueprint vs Running instance

Image:

- Read-only template
- Like a Java class
- Built from Dockerfile

Container:

- Running instance of image
- Like a Java object
- Has its own process, network, storage

One image → many containers

💡 Image = recipe. Container = cooked dish.



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Dockerfile — Your Recipe

5 lines to containerize Spring Boot

```
FROM openjdk:17-slim
WORKDIR /app
COPY target/app.jar app.jar
EXPOSE 8080
CMD ["java", "-jar", "app.jar"]
```

- Start from Java 17 base
- Copy your JAR file
- Tell Docker which port
- Run the app

💡 This is all you need for Spring Boot



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linkedin.com/in/bhatishubham

Docker Compose — Multi-Service

App + DB + Cache in one command

```
services:
  app:
    build: .
    ports: ["8080:8080"]
    depends_on: [db, redis]
  db:
    image: mysql:8
    environment:
      MYSQL_ROOT_PASSWORD: secret
  redis:
    image: redis:7
```

- docker-compose up
- Everything starts together

💡 docker-compose up = entire system running



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linkedin.com/in/bhatishubham

Essential Commands

The only 8 you need

```
docker build -t myapp .  
docker run -p 8080:8080 myapp  
docker ps          # running containers  
docker logs       # see output  
docker stop       # stop container  
docker images     # list images  
docker-compose up  # start all  
docker-compose down # stop all
```

💡 Master these 8. That covers 95% of daily use.



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Docker in Interviews

Questions they'll ask

Q: Image vs Container?

A: Blueprint vs running instance

Q: How to reduce image size?

A: Multi-stage builds + slim base

Q: How services communicate?

A: Docker network + service names

Q: Data persistence?

A: Volumes (docker volume create)

💡 Docker is asked in every backend interview now



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linkedin.com/in/bhatishubham

done = true

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Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)



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